

Example 8: It can be verified that the matrix

$$A = [a_1 \ a_2 \ \dots \ a_5] = \begin{bmatrix} 1 & 3 & 3 & 2 & -9 \\ -2 & -2 & 2 & -8 & 2 \\ 2 & 3 & 0 & 7 & 1 \\ 3 & 4 & -1 & 11 & -8 \end{bmatrix}$$

is row equivalent to the matrix B in Example 7. Find a basis for $\text{Col } A$.

Solution: From Example 7, the pivot columns of A are columns 1, 2 and 5. Also, $b_3 = -3b_1 + 2b_2$ and $b_4 = 5b_1 - b_2$.

Since row operations do not affect linear dependence relations among the columns of the matrix, we claim that $a_3 = -3a_1 + 2a_2$ and $a_4 = 5a_1 - a_2$.

$$a_3 = \begin{bmatrix} 3 \\ 2 \\ 0 \\ -1 \end{bmatrix} \quad -3a_1 + 2a_2 = -3 \begin{bmatrix} 1 \\ -2 \\ 2 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} 3 \\ -2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -3 \\ 6 \\ -6 \\ -9 \end{bmatrix} + \begin{bmatrix} 6 \\ -4 \\ 6 \\ 8 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 0 \\ -1 \end{bmatrix}$$

$$\therefore a_3 = -3a_1 + 2a_2$$

$$a_4 = \begin{bmatrix} 2 \\ -8 \\ 7 \\ 11 \end{bmatrix} \quad 5a_1 - a_2 = 5 \begin{bmatrix} 1 \\ -2 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 3 \\ -2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 5 \\ -10 \\ 10 \\ 15 \end{bmatrix} - \begin{bmatrix} 3 \\ -2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ -8 \\ 7 \\ 11 \end{bmatrix}$$

$$\therefore a_4 = 5a_1 - a_2$$

By the argument in Example 7, a_3 and a_4 are not needed to generate the column space of A. Also, $\{a_1, a_2, a_5\}$ must be linearly independent, because any dependence relation among a_1, a_2 and a_5 would imply the same dependence relation among b_1, b_2 and b_5 . Since $\{b_1, b_2, b_5\}$ is linearly independent, $\{a_1, a_2, a_5\}$ is also linearly independent and hence is a basis for $\text{Col } A$. $\square$

The argument in Example 8 can be adapted to prove the following theorem:

Theorem 13: The pivot columns of a matrix A form a basis for the column space of A.